VectorQuay
INDIGO SIGNAL

PRODUCT REQUIREMENTS DOCUMENT

Phase 0 PRD

Objective, feasibility, and trading constraints for the first go/no-go decision.

VERSION	STATUS
v1.0	**In Review**

OWNER	THEME
Brad Malia	**Indigo Signal**

Executive Summary

Phase 0 exists to determine whether VectorQuay should proceed into implementation. It converts the approved roadmap into a product framing exercise that validates feasibility, clarifies restrictions, and defines the conditions required before any engineering work moves into trading system implementation.

Phase Goal

Prove that the product objective, exchange constraints, initial capital, and operator rules are coherent enough to justify building the platform.

Purpose

Phase 0 establishes whether VectorQuay should proceed into implementation. This phase exists to validate that the product goal, starting capital, exchange constraints, and risk boundaries are coherent enough to justify building the system.

Product Goal

VectorQuay aims to maximize fee-adjusted portfolio growth as fast as possible through informed autonomous trading on Coinbase spot markets, while remaining inside hard safety, capital, and operational limits.

Background

The product will begin with approximately $50 in existing Coinbase-held ETH and may autonomously rebalance that capital into other Coinbase-listed spot assets. The bot must not move funds to or from the user's bank account in v1. The product is intended to run locally, provide a GUI administration experience, and remain suitable for future resale through bring-your-own-credentials design.

In Scope

- Confirm exact product boundaries and prohibited actions.
- Validate Coinbase spot-only feasibility for the starting account conditions.
- Validate fee, minimum order size, and tradable-asset constraints at approximately $50 initial capital.
- Define tradable-pair viability rules, including quote-currency restrictions, liquidity thresholds, spread tolerances, and minimum-order compatibility.
- Define measurable objectives, risk boundaries, and aggressiveness modes.
- Define the initial asset-allowlist policy for v1 planning.
- Establish go/no-go criteria for implementation.

Out of Scope

- Building trading logic.
- Connecting to Coinbase APIs beyond research and feasibility analysis.
- Implementing AI-driven decisioning.
- Implementing paper trading or live trading.
- Automating bank deposits or withdrawals.
- Supporting non-Coinbase exchanges.

Users and Stakeholders

- Primary operator and owner: Brad Malia.
- Future potential customers of a resale-ready version.
- Future engineering and operator users who will need clear documentation, controls, and recoverability.

Product Constraints

- Coinbase spot only for v1.
- Bot may trade only Coinbase-listed assets.
- Bot is a trade-only account actor for v1.
- Bot may use only funds already available in the Coinbase account.
- No bank withdrawals to add funds.
- No bank deposits out of Coinbase for v1.
- The bot may not add, modify, or remove bank accounts or payment methods.
- The bot may not transfer assets to external wallets or external destinations.
- Initial capital is approximately $50, currently in ETH.
- The bot may sell ETH and rotate capital into other approved assets.
- Asset-to-asset rotation must occur only through approved quote-currency trading flows.
- For v1, approved quote currencies must be explicitly limited to USD and USDC unless Phase 0 feasibility findings narrow the policy further.
- A human operator may designate assets as protected or leave-alone assets with explicit protection modes.
- For v1, portfolio valuation, reporting, and benchmark comparisons must use USD as the reference currency.
- For v1, USDC is allowed both as an approved routing asset and as an optional defensive capital-allocation state.
- The application must run on Ubuntu and maintain a compatibility path for Windows 11.
- Admin controls must be exposed through a GUI application.
- No user-specific secrets may be stored in the repository.

Allowed and Prohibited Account Actions

Allowed in v1

- Read balances, orders, fills, supported products, and market data.
- Place Coinbase spot buy and sell orders.
- Cancel bot-owned open orders only.
- Reconcile balances, positions, and order outcomes.
- Rotate from one crypto asset into another through approved quote-currency trading flows.
- Respect operator-designated protected or leave-alone assets according to their configured protection mode.

Prohibited in v1

- Bank deposits, bank withdrawals, or payment-method changes, including adding new bank accounts.
- Crypto withdrawals, wallet transfers, or transfers to any external wallet or external destination.
- Staking, Web3 interactions, or any on-chain wallet operations.
- Margin, borrowing, derivatives, futures, or perpetuals.
- Recurring buys, non-strategy conversion flows, or profit-sweep automation.
- Cancelling or modifying human-owned orders without explicit human approval.
- Violating the configured protection mode of any operator-designated protected or leave-alone asset.
- Any account action unrelated to market data, balances, orders, fills, or risk enforcement.

Permissions Matrix

Action	Status	Notes
Read balances	Allowed	Required for account awareness and risk checks.
Read orders and fills	Allowed	Required for reconciliation and order-state tracking.
Read supported products and market data	Allowed	Required for decisioning and feasibility validation.
Place spot buy orders	Allowed	Only within approved strategy, risk, and asset rules.
Place spot sell orders	Allowed	Only within approved strategy, risk, and asset rules.
Cancel bot-owned open orders	Allowed	Only for orders positively identified as bot-owned.
Cancel human-owned open orders	Allowed with human approval only	Default behavior is prohibited without explicit operator approval.
Rotate one crypto asset into another	Allowed	Must occur only through approved quote-currency flows, limited to USD and USDC for v1 unless narrowed.
Trade protected asset in no-buy mode	Partially allowed	Bot may sell or hold, but may not buy additional quantity.
Trade protected asset in no-sell mode	Partially allowed	Bot may buy or hold, but may not sell existing quantity.
Trade protected asset in fully excluded mode	Prohibited	Bot may neither buy nor sell the asset.
Bank deposits	Prohibited	No automatic funding from bank accounts.

Action	Status	Notes
Bank withdrawals	Prohibited	No automatic transfer of funds out to bank accounts.
Add, modify, or remove bank accounts/payment methods	Prohibited	Explicitly disallowed in v1.
Transfer assets to external wallet or destination	Prohibited	Explicitly disallowed in v1.
Staking	Prohibited	Outside trade-only account policy.
Web3 or on-chain interactions	Prohibited	Outside trade-only account policy.
Margin, borrowing, derivatives, futures, or perpetuals	Prohibited	Spot-only v1 policy.
Recurring buys or profit-sweep automation	Prohibited	Outside approved strategy engine and deferred from v1.

Operational Control Rules

- Bot-owned orders must be identified through bot-generated order metadata such as client order identifiers or equivalent markers.
- Any order not carrying a bot-owned identifier must be treated as human-owned by default.
- Protected assets must support explicit operator-selected modes: no buy, no sell, or fully excluded.
- If the bot detects unexpected human trading activity or unknown active orders, Phase 0 must define whether the correct v1 behavior is pause, restricted mode, or continue-with-ignore before implementation begins.
- Phase 0 must produce a clear operator override policy for mixed human/bot account activity.

Functional Requirements

- Document the explicit business objective and how success will be measured.
- Document hard restrictions on account and funding behavior.
- Document the explicit allowed and prohibited account-action matrix for v1.
- Document the rule that bot-owned orders may be cancelled by the bot, but human-owned orders may not be cancelled without explicit human approval.
- Document the operator-controlled protected-asset policy so selected assets can be marked as no buy, no sell, or fully excluded.
- Document the order-ownership classification rule so bot-owned and human-owned orders can be distinguished deterministically.
- Document required bot behavior when unexpected human activity or unknown orders are detected.
- Determine whether Coinbase rules and economics permit meaningful trading at the starting capital level.
- Define initial quote-currency policy and pair-routing rules, with v1 limited to USD and USDC unless feasibility findings require a narrower scope.
- Define minimum liquidity, maximum spread, and minimum-order compatibility rules for the initial tradable universe.
- Define whether USDC is permitted as a strategic defensive holding in addition to being a routing asset.
- Define aggressiveness profiles for future strategy work: conservative, balanced, and aggressive.
- Define the initial asset-allowlist policy for v1 planning.
- Produce a mandatory end-of-phase decision outcome with supporting rationale: proceed as planned, proceed with narrower scope, proceed with higher required capital, or stop.

Non-Functional Requirements

- All findings must be documented in version-controlled project documentation.
- All constraints must be written clearly enough to become future acceptance criteria.
- The output must be understandable without relying on chat transcript history.
- Documentation must support future resale and multi-user productization decisions.

Success Metrics

- Primary metric defined: fee-adjusted account growth.
- Secondary metrics defined: drawdown, turnover, slippage sensitivity, and growth versus passive hold.
- Initial aggressiveness modes documented.
- Coinbase feasibility conclusion reached for the starting capital level.
- Tradable-pair viability rules are defined for the initial asset universe.
- Explicit account-action permission matrix is approved.
- Order ownership classification and mixed human/bot control behavior are defined.
- Protected-asset semantics are defined as no buy, no sell, or fully excluded.
- USD is confirmed as the v1 reporting and valuation currency.
- A mandatory end-of-phase decision outcome is produced and approved.

Key Risks to Resolve

Phase 0 Risks

- The starting capital may be too small for profitable or meaningful execution after fees.
- Coinbase minimum order sizes may make certain strategies impossible.
- Frequent trading may destroy returns at low capital due to fees and spread.
- “Maximize profits quickly” may create pressure toward unsafe aggressiveness without explicit controls.
- Future resale goals may be compromised if the early design becomes too single-user or environment-specific.

Deliverables

- Approved Phase 0 PRD.
- Phase 0 technical requirements document.
- Phase 0 unit test plan.
- Phase 0 integration test plan.
- Phase 0 regression test plan.
- Documented feasibility findings and go/no-go criteria.
- Documented account-action permission matrix.
- Documented initial quote-currency and tradable-pair viability policy.
- Documented order-ownership and mixed-control policy.
- Documented protected-asset mode policy.
- Documented end-of-phase decision outcome.

Open Decisions

- The exact initial v1 asset allowlist still needs to be finalized.
- The exact quantitative thresholds for acceptable drawdown and turnover still need to be set.
- The minimum capital level for a viable live rollout may need revision after feasibility analysis.

Acceptance Criteria

Approval Conditions

- The product objective is explicitly defined and approved.
- Hard trading and funding restrictions are documented and approved.
- The v1 allowed and prohibited account-action matrix is documented and approved.
- The initial quote-currency and tradable-pair viability policy is documented.
- The order-ownership model and mixed human/bot activity policy are documented.
- The protected-asset mode policy is documented.
- Coinbase feasibility questions for the initial capital level are identified clearly enough to investigate in the technical requirements phase.
- Initial aggressiveness modes are defined.
- Go/no-go decision criteria and required end-of-phase decision outcomes are documented.
- The document is suitable as the baseline input to the Phase 0 technical requirements document.

Recommendation

Proceed to the Phase 0 technical requirements document. That document should convert this approved product intent into a concrete research, architecture, feasibility, documentation, and test planning package.